Feedback:
- User requirements: B-: Reasonable/pootwritten: check some are not well specified, ambiguous, not meaningful/verifiable, traceable, use of shall/should.
- System requirements: B+: Reasonable/written, check some should be detailing, correctly categorised, traceable not detailed enough to capture system view, many ambiguous, not meaningful/verifiable, poor use of shall/should for some
- Effort estimation: B+: Reasonable/, need to show full calculation and make full use of all developers?

Overall Evaluation: B: needs major improvements

erifiabl

ningful

Department of Electrical and Computer Engineering

Software Engineering – COMP 433 | Second Semester 2025/2026

Phase 2: Requirements Engineering

Elicitation & Discovery

Business: Cooker Shop

Group 7 — Developer Group

#	Name	Student ID
1	Asmaa Jaraba	1231081
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5	Abdel Rahman Sabbagh	

Instructor: Dr. Adel Taweel

Section: 2

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B- Task 2.1 — User Requirements

avoid system design decisions

Lead: Asmaa Jaraba (UR1, UR2), Masa Awad (UR3, UR4), Anwar Atawna (UR5, UR6), Abd[illegible] Rahman Sabbagh (UR7), Misk Mashahreh (UR8) **Contributors:** *Asmaa Jaraba & Misk Mashahreh (finalization & discussion & reviewing))*

UR1 — Product Browsing: The system shall enable customers with an online product catalog displaying all available cooker products.

Unnecessary confusing details for UR - these details should be specified at SR level

UR2 — Product Search: The system shall enable customers to search for cookers by name, brand, or type.

Not meaningful/ ambiguous/not verifiable functional requirement

UR3 — Shopping Cart: The system shall enable customers to add selected [illegible] shopping cart.

Not meaningful functional term use for requirement

UR4 — User Account M[illegible] shall enable customers to register an account, log in, and maintain their perso[illegible] and saved delivery details.

ambiguous UR, two/several verbs or functions?

UR5 — Online Ordering: The system sha[illegible]able customers to place and confirm online orders.

UR6 — Delivery & Installation Scheduling: Th[illegible] system shall enable custom[illegible]rs to schedule delivery and installation appointments for their purchased cookers and receive bookin[illegible] confir[illegible]ations.

UR7 — Product Management: The system shall allow the store administrator to [illegible]nag[illegible] Add, Update, Delete) the cooker product catalog.

avoid a system level function

UR8 — Maintenance Requests: The system shall enable customers to submit maintena[illegible] requests for cookers purchased from the store.

ambiguous UR, two/several verbs or functions?

A user requirement is one that defines an action, an ability that needs to be provided to serve its users as part of the business services. Each user requirement is unique, providing a unique business service.

Some possible key/correct URs are missing:
- Order tracking
- Order Cancelation
- payment
- etc?

System requirement is one that defines an action, an ability that need to provided to serve its users, detailing specific needs of ONE Specific User Requirement, to support the respective business services

B+

Task 2.2 — System Requirements

Lead: Asmaa Jaraba (SR1, SR2), Masa Awad (SR3, SR4), Anwar Atawna (SR5, SR6), Abdel Rahman Sabbagh (SR7), Misk Mashahreh (SR8) **Contributors:** *Asmaa Jaraba & Misk Mashahreh (finalization & discussion & reviewing))*

SR1 — Product Browsing (derived from UR1)

Avoid GUI specs

SR1.1: The system shall display a catalog of all available cooker products on a dedicated Products page, accessible from the main navigation menu.

SR1.2: Each product listing shall present the product name, brand, cooker type (gas, electric, induction, or hybrid), price, and a summary of key features.

SR1.3: The system shall display a stock status label (In Stock / Out of Stock) on each product listing.

SR1.4: The system shall present a full product detail page, containing additional images and complete technical specifications, when a customer selects a product.

SR1.5: The system shall provide a category filter panel enabling customers to filter the product catalog by cooker type.

SR1.6: The system shall support pagination or continuous scrolling when the catalog contains more than twenty products per page.

SR2 — Product Search (derived from UR2)

Avoid GUI specs

SR2.1: The system shall provide a search bar enabling customers to search the product catalog by product name, brand, or cooker type.

SR2.2: The system shall display all matching search results after the customer submits a search query.

SR2.3: The system shall display a 'No product found' message when no products match the entered search term.

SR2.4: The system shall enable customers to select between two and four products for simultaneous comparison.

SR2.5: The system shall display a side-by-side comparison table for selected products, showing name, brand, type, price, and key features.

SR2.6: The system shall enable customers to remove a product from the comparison and replace it with another, without clearing the entire comparison.

SR3 — Shopping Cart (derived from UR3)

SR3.1: The system shall enable customers to add one or more cooker products to a shopping cart from the product catalog or product detail page.

SR3.2: The system shall display cart contents showing product name, unit price, quantity, and line total for each item.

SR3.3: The system shall enable customers to increase or decrease the quantity of any item in the cart.

SR3.4: The system shall enable customers to remove any individual product from the cart.

SR3.5: The system shall automatically recalculate and display the cart total price whenever the cart contents change.

SR3.6: The system shall provide a checkout button enabling customers to proceed from the cart to the order confirmation step.

Avoid GUI specs

SR4 — User Account Management (derived from UR4)

what happens when customer forgets password?
what happens after multiple incorrect attempts?

SR4.1: The system shall register a new customer account upon submission of a full name, a valid email address, and a password of at least eight characters.

SR4.2: The system shall validate that the registration email is correctly formatted and is not already associated with an existing account.

SR4.3: The system shall authenticate registered customers upon submission of a matching email address and password.

SR4.4: The system shall enable logged-in customers to update personal information including name, email address, phone number, and delivery address.

SR4.5: The system shall terminate a customer session and log the customer out securely upon request.

SR5 — Online Ordering (derived from UR5)

SR5.1: The system shall require customers to be authenticated before processing an order.

SR5.2: The system shall validate all required order fields — including delivery address, phone number, and payment details — before accepting a submission.

Not meaningful/ ambiguous/not verifiable functional requirement

SR5.3: The system shall reject orders for products marked as Out of Stock and display an appropriate notification “Product X Out of Stock” to the customer.

SR5.4: The system shall present a complete order summary — including product details, quantities, total cost, and delivery address — for customer review prior to confirmation.

SR5.5: The system shall process secure online payments by credit and debit card through an integrated secure payment gateway. Not meaningful/ ambiguous/not verifiable functional requirement

SR5.6: The system shall generate a unique order identifier and dispatch an order confirmation to the customer’s registered email address upon successful placement.

SR6 — Delivery & Installation Scheduling (derived from UR6)

SR6.1: The system shall present available delivery time slots based on the customer’s delivery location and current delivery team workload.

SR6.2: The system shall enable customers to book an installation appointment either simultaneously with, or independently from, the delivery booking.

SR6.3: The system shall dispatch an automated email confirmation to the customer once a delivery or installation slot is successfully booked.

SR6.4: The system shall prevent double-booking of the same delivery personnel or installation technician for overlapping time slots.

SR6.5: The system shall enable customers to reschedule or cancel a booking at least 24 hours before the scheduled time.

SR7 — Product & Inventory Management (derived from UR7)

SR7.1: The system shall enable the administrator to add new cooker products to the catalog, specifying name, brand, type, size, price, features, images, and availability status.

SR7.2: The system shall enable the administrator to update any existing product details including name, description, price, images, and specifications.

SR7.3: The system shall enable the administrator to deactivate or permanently remove a product from the catalog.

SR7.4: The system shall enable the administrator to set and update each product's stock status as In Stock or Out of Stock.

SR7.5: The system shall reject any customer attempt to add or purchase a product marked as Out of Stock.

SR7.6: The system shall provide the administrator with search and filter functionality in the management dashboard, by product name, category, brand, or stock status.

SR8 — Maintenance Requests (derived from UR8)

SR8.1: The system shall accept a maintenance request from a customer upon submission of a valid invoice number, product model number, and a written description of the reported issue.

SR8.2: The system shall verify that the submitted invoice number corresponds to a purchase recorded in the system before processing the maintenance request.

SR8.3: The system shall check whether the product is within its warranty period based on the purchase date associated with the invoice number.

SR8.4: The system shall require customers to attach at least one image or video file illustrating the reported malfunction before submission.

SR8.5: The system shall generate a unique maintenance ticket identifier and send an acknowledgement email to the customer confirming receipt of the maintenance request.

B+

Task 2.3 — Effort, Time & Cost Estimation

Lead: Misk Mashahreh **Contributors:** *Asmaa Jaraba (reviewing); Asmaa Jaraba & Misk Mashahreh (finalization)*

UR	Estimated Effort	Estimated No of Developers	Total Effort
UR1	2 pw	1	= 2 x 1 = 2 pw
UR2	2 pw	1	= 2 x 1 = 2 pw
UR3	3 pw	2	= 3 x 2 = 6 pw
UR4	3 pw	2	= 3 x 2 = 6 pw
UR5	4 pw	2	= 4 x 2 = 8 pw
UR6	3 pw	2	= 3 x 2 = 6 pw
UR7	3 pw	1	= 3 x 1 = 3 pw
UR8	2 pw	1	= 2 x 1 = 2 pw
Total effort/avg	**22 pw**	(1+1+2+2+2+2+1+1)/8 = 1.5 dev on avg needed	**35 pw**
Schedule time 30%	22 x 1.30 = 28.6 ≈ 29w (min time to complete)		35 x 1.30 = 45.5 ≈ 46w (max time to complete)
Cost		**Avg salary = $400**	$400 x 46w = $18,400
Profit margin (min=10%; max=30%)		**Min cost →** **Max cost →**	$18,400 x 1.10 = $20,240 $18,400 x 1.30 = $23,920

How many developers in your team?
Did you think of utilising all of them to optimise development time?
Development time longer than the semester time?

Minutes of Meetings

Lead: *Asmaa Jaraba (note-taking & review); Asmaa Jaraba & Misk Mashahreh (finalization); all members (attendance & discussion)*

Meeting 1	
Date	Saturday, 2 May 2026
Time	7:00 PM – 7:30 PM
Location	Online (Video Call)
Developer Group	Asmaa Jaraba Misk Mashahreh Masa Awad Anwar Atawna Abdel Rahman Sabbagh

Meeting 1	
Customer Group	Samir Ali Ibrahim Al Ardah Maen Foqaha Yahya Sarhan
Topics Discussed	1. General overview of the Cooker Shop business, its current manual operations, main problems, and digital goals. 2. Discussion of the core services the system must support: product browsing, ordering, payment, delivery, and installation. 3. Clarification of the four cooker types sold (gas, electric, induction, hybrid) and the approximately 60 product models available. 4. Agreement on the need for customer accounts to register, log in, track orders, and manage personal and delivery details. 5. Identification of administrator functions: product catalog management, stock control, and order processing. 6. Discussion of after-sales support: the customer group mentioned warranty-based maintenance but did not initially provide full details. 7. UR8 (After-Sales Maintenance Requests) was identified as unclear in the original business outline provided in Phase 1. The developer group queried whether maintenance should be handled through the online system. The customer group confirmed that an online maintenance request feature is required, with warranty verification performed via the customer's invoice number. This was agreed upon and formalised as UR8.

Meeting 2	
Date	Sunday, 3 May 2026
Time	9:00 AM – 9:20 AM
Location	Birzeit UNI
Developer Group	Asmaa Jaraba Misk Mashahreh Masa Awad Anwar Atawna Abdel Rahman Sabbagh
Customer Group	Samir Ali Ibrahim Al Ardah Maen Foqaha Yahya Sarhan
Topics Discussed	1. Full review of the drafted User Requirements (UR1–UR8) with the customer group. 2. Customer group confirmed all eight user requirements accurately reflect their business needs and services. 3. Clarification that installation scheduling may be booked simultaneously with, or independently from, delivery. 4. Confirmation that maintenance requests must be tied to a valid invoice number and subject to warranty period verification.

Meeting 2	
	5. Payment methods confirmed: credit/debit card online payment is required; cash on delivery is a desirable option. 6. Admin dashboard scope agreed: product management, order processing, and stock status control. 7. Final validation and sign-off on all eight requirements before proceeding to system requirements documentation.

Did you include your own team meet? keep record of all meetings